

## EXPERIMENT 12:

### AIM:

To implement an Earley parser for a context-free grammar using Python.

### PROGRAM:

```
from nltk import CFG
from nltk.parse import EarleyChartParser

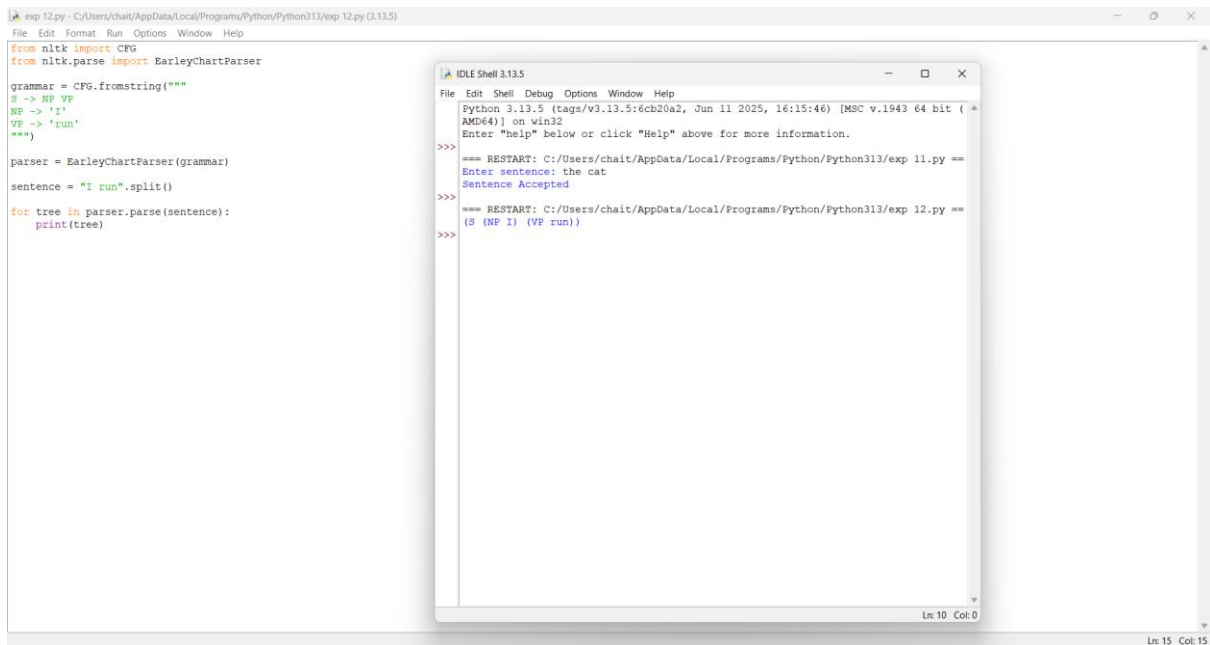
grammar = CFG.fromstring("""
S -> NP VP
NP -> 'I'
VP -> 'run'
""")

parser = EarleyChartParser(grammar)

sentence = "I run".split()

for tree in parser.parse(sentence):
    print(tree)
```

### OUTPUT:



The screenshot shows a Python IDE window titled 'exp 12.py' with the following code:

```
from nltk import CFG
from nltk.parse import EarleyChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> 'I'
VP -> 'run'
""")

parser = EarleyChartParser(grammar)

sentence = "I run".split()

for tree in parser.parse(sentence):
    print(tree)
```

Overlaid on this is a smaller window titled 'IDLE Shell 3.13.5' showing the execution output:

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
=== RESTART: C:/Users/chaith/AppData/Local/Programs/Python/Python313/exp 11.py ===
Enter sentence: the cat
Sentence Accepted

>>>
=== RESTART: C:/Users/chaith/AppData/Local/Programs/Python/Python313/exp 12.py ===
(S (NP I) (VP run))

>>>
```

The status bar at the bottom right of the IDE shows 'Ln: 15 Col: 15'.